Pure Mathematics with Mechanics GCE A Level 2022 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Multiple choice paper (0765 Pure Maths with Mechanics 1, June 2022, Advanced Level, one and a half hours). Answer ALL questions; each question has FOUR suggested answers A, B, C and D and only one answer must be marked (if more than one answer is marked the score for that question is zero). Formulae booklets and calculators are allowed. Section A: Pure Mathematics (questions 1 to 35); Section B: Mechanics (questions 36 to 50). Mobile phones are NOT allowed in the examination room.

SECTION A: PURE MATHEMATICS

The remainder when the polynomial $f(x) = 2x^3 + ax^2 - 3x + 1$ is divided by $(x + 1)$ is -2. The value of the constant a is

a) 3
b) 4
c) -5
d) -4

2. If the roots of the equation $2x^2 + 5x - 3 = 0$ are α and β, then the value of $\alpha^2 + \beta^2$ is

a) $\frac{29}{4}$
b) $\frac{37}{4}$
c) $\frac{25}{4}$
d) $\frac{31}{4}$

3. If $\frac{x+1}{(x-3)(x-5)} \equiv \frac{M}{x-3} + \frac{N}{x-5}$, then

a) $M = 2$ and $N = 3$
b) $M = -2$ and $N = 3$
c) $M = 3$ and $N = 2$
d) $M = -2$ and $N = -3$

4. The solution set of the inequality $|2x - 1| > 3$ is

a) $\{x : x < -1 \text{ or } x > 2\}$
b) $\{x : x < -2 \text{ or } x > 1\}$
c) $\{x:-1 d) $\{x:-2

5. A partial order relation is that which is

a) reflexive, symmetric and transitive
b) reflexive and anti-symmetric
c) reflexive, anti-symmetric and transitive
d) anti-symmetric and transitive

6. The number of distinct arrangements of the letters of the word ABRACADABRAR is

a) $\frac{12!}{5!\,3!\,2!}$
b) $\frac{12!}{5!\,3!\,3!}$
c) $\frac{12!}{5!\,2!\,2!}$
d) $\frac{12!}{5!\,4!\,3!}$

7. A function $f : \mathbb{R} \to \mathbb{R}$ is given by $f(x) = ax + b$. If $f(3) = 3$ and $f(4) = 5$, then the values of the constants a and b are

a) $a = 2,\ b = 3$
b) $a = -2,\ b = 3$
c) $a = 2,\ b = -3$
d) $a = -2,\ b = -3$

8. The value of $2\log_4 8$ is

a) 2
b) $\frac{3}{2}$
c) 3
d) $\frac{2}{3}$

9. If $u_n = 5 + \frac{3}{2}(n-1)$ is the nth term of an arithmetic progression, the sum of the first twenty terms of the progression is

a) 770
b) 385
c) 358
d) 538

10. The solution set of the inequality $\frac{(1-x)(x-3)}{(x+2)} > 0$ is

a) $\{x:x<-2\ \cup\ 1 b) $\{x:-23\}$
c) $\{x:x<1\ \cup\ 2 d) $\{x : x < -2\ \cup\ x > 3\}$

11. p and q are two statements. The compound statement $p \Rightarrow q$ is false when

a) p and q have the same truth value
b) p and q are converse of each other
c) p is true and q is false
d) both p and q are false

12. If $a + bi = \dfrac{1+i}{1-2i}$, the values of the constants a and b are

a) $a = \dfrac{1}{5},\ b = \dfrac{3}{5}$
b) $a = \dfrac{1}{5},\ b = -\dfrac{3}{5}$
c) $a = -\dfrac{1}{5},\ b = \dfrac{3}{5}$
d) $a = -\dfrac{1}{5},\ b = -\dfrac{3}{5}$

13. If $x^2 - 4x + 9 \equiv (x + A)^2 + B$, then

a) $A = 2,\ B = 3$
b) $A = 2,\ B = 5$
c) $A = -2,\ B = 3$
d) $A = -2,\ B = 5$

14. Two variables y and x are connected by the equation $y = a(x+2)^k$. A linear equivalence of this equation is

a) $\log y = \log k(x+2) + \log a$
b) $\log y = a\log(x+2) + \log k$
c) $\log y = k\log 2x + \log a$
d) $\log y = k\log(x+2) + \log a$

15. If matrix $\mathbf{A} = \begin{pmatrix} -1 & k & 2 \\ 2 & -6 & 0 \\ 1 & -2 & -4 \end{pmatrix}$, then the value of k for which $|\mathbf{A}| = 4$ is

a) 2
b) 3
c) -3
d) -2

16. If $f : \mathbb{R} \to \mathbb{R}$, where $f(x) = \ln(4 + x^2)$, then $f'(2) =$

a) $\dfrac{1}{4}$
b) 2
c) 4
d) $\dfrac{1}{2}$

17. The period of the trigonometric function $f : \mathbb{R} \to \mathbb{R}$, where $f(x) = \sin\dfrac{2}{3}x$ is

Options as printed: A) $\dfrac{4}{3}\pi$ B) $\dfrac{2}{3}\pi$ C) $\dfrac{3}{4}\pi$ D) $\dfrac{3}{2}\pi$

18. If a polynomial $P(x)$ produces a quotient $(x^2 + 2x + 1)$ and a remainder 5 when divided by $(x+1)$, then $P(x) =$

a) $x^3 + 3x^2 + 3x + 6$
b) $x^3 + 3x^2 + 4x + 6$
c) $x^3 + 4x^2 + 3x + 6$
d) $x^3 + 2x^2 + 3x + 6$

19. The parametric equations of a curve are $1 - x = \tan\theta$ and $[y] = \dfrac{1}{\cos\theta}$. The cartesian equation of this curve is

a) $x^2 - y^2 + 2x + 2 = 0$
b) $x^2 - y^2 - x + 2 = 0$
c) $x^2 - y^2 - 2x + 2 = 0$
d) $x^2 - y^2 + x + 2 = 0$

20. If the cartesian equation of a plane is $x - 2y + z - 5 = 0$, then the vector normal to this plane is

Options as printed: A) $\mathbf{i} - 2\mathbf{j} - \mathbf{k}$ B) $\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ C) $\mathbf{i} - 2\mathbf{j} - \mathbf{k}$ D) $\mathbf{i} + 2\mathbf{j} + \mathbf{k}$

21. Some values of the function $f : \mathbb{R} \to \mathbb{R}$ are given in the table below:

x	2	3	4	5
$f(x)$	5	15	28	k

If by the trapezium rule $\displaystyle\int_2^5 f(x)\,dx \approx 68$, then the value of the constant k is

a) 48
b) 42
c) 47
d) 45

22. If x_1 is a first approximate root of the equation $f(x) = 0$. By Newton-Raphson's procedure, a second approximate root of this equation is given by

a) $x_1 - \dfrac{f'(x_1)}{f(x_1)}$
b) $x_1 + \dfrac{f'(x_1)}{f(x_1)}$
c) $x_1 - \dfrac{f(x_1)}{f'(x_1)}$
d) $x_1 + \dfrac{f(x_1)}{f'(x_1)}$

23. A function f is defined by $\mathrm{f}(x) = \dfrac{2x - 3}{x + 1}$, $x \in \mathbb{R}$, $x \neq -1$, the value of $\mathrm{f}^{-1}(0)$ is

a) $-\dfrac{3}{2}$
b) $\dfrac{3}{2}$
c) $\dfrac{2}{3}$
d) $-\dfrac{2}{3}$

24. The values of λ and μ for which the lines $\mathbf{r} = 3\mathbf{i} - \mathbf{j} + \mathbf{k} + \lambda(\mathbf{i} + 2\mathbf{j} - \mathbf{k})$ and $\mathbf{r} = 4\mathbf{i} + 4\mathbf{j} + \mathbf{k} + \mu(-\mathbf{i} + \mathbf{j} + 2\mathbf{k})$ intersect are

a) $\lambda = 2$ and $\mu = 1$
b) $\lambda = 2$ and $\mu = -1$

c) $\lambda = -2$ and $\mu = -1$
d) $\lambda = -2$ and $\mu = 1$

25. The number of ways of arranging 5 girls on a bench if the two left-handed girls must be together is

a) 24
b) 48
c) 60
d) 96

26. $\sin\left(x + \frac{\pi}{6}\right) - \sin\left(x - \frac{\pi}{6}\right) =$

a) $-\cos x$
b) $\sin x$
c) $\cos x$
d) $-\sin x$

27. The graph of $y = \dfrac{4}{x^2 - 1}$ has

a) two vertical asymptotes $x = \pm 1$ and no horizontal asymptote
b) one vertical asymptote $x = 1$ and one horizontal asymptote $y = 0$
c) two vertical asymptotes $x = \pm 1$ and a horizontal asymptote $y = 4$
d) two vertical asymptotes $x = \pm 1$ and a horizontal asymptote $y = 0$

28. Given the differential equation $\dfrac{dy}{dx} - 2xy = 0$. If $y = 1$ when $x = 1$, then

a) $y = e^{x^2 - 1}$
b) $y = e^{x^2 + 1}$
c) $y = e^{x^2}$
d) $y = e^{1 - x^2}$

29. The coefficient of x^2 in the binomial expansion of $\left(1 + \frac{1}{2}x\right)^{\frac{1}{2}}$ is

a) $\dfrac{1}{32}$
b) $-\dfrac{1}{16}$
c) $\dfrac{1}{16}$
d) $-\dfrac{1}{32}$

30. The coordinates of the points where the curves $y = 2x - 3$ and $y = x^2 - 2x$ intersect are

a) $(1, -1)$ and $(3, 3)$
b) $(1, 1)$ and $(3, 2)$
c) $(-1, -1)$ and $(3, 3)$
d) $(1, -1)$ and $(3, 2)$

31. The value of the constant k for which $\displaystyle\sum_{r=1}^{\infty} \left(\frac{[k]}{2^{r-1}}\right) = 10$ is

a) $\frac{1}{5}$
b) 5
c) 2
d) $\frac{1}{2}$

32. If a linear reduction of the relationship $y = ax^n$ produced the result $\log y = 2\log x + 1.3$, then the approximate values of a and n are

a) $a = 2$ and $n = 2$
b) $a = 20$ and $n = 2$
c) $a = 2$ and $n = 20$
d) $a = 20$ and $n = 20$

33. The complex number z, in the form $a + bi$, for which $(1 - 2i)z = 3 - i$ is

a) $1 - i$
b) $-1 - i$
c) $1 + i$
d) $-1 + i$

34. The value of a for which the quadratic equation $2x^2 - 6x + 3a = 0$ has equal roots is

a) $\frac{2}{3}$
b) $-\frac{2}{3}$
c) $-\frac{3}{2}$
d) $\frac{3}{2}$

35. $\int_1^2 \frac{x}{1 + 2x^2}\,dx =$

a) $4\ln 3$
b) $\frac{1}{2}\ln 3$
c) $2\ln 3$
d) $\frac{1}{4}\ln 3$

SECTION B: MECHANICS

A particle has position vector $\mathbf{r}$ at time t seconds given by $\mathbf{r} = \left(e^{2t}\mathbf{i} + e^{3t}\mathbf{j}\right)$ m. The velocity and acceleration of the particle when $t = 0$ are respectively

a) $(2\mathbf{i} + 3\mathbf{j})\ \mathrm{m\,s^{-1}},\ (4\mathbf{i} + 9\mathbf{j})\ \mathrm{m\,s^{-2}}$
b) $(2\mathbf{i} + 3\mathbf{j})\ \mathrm{m\,s^{-1}},\ (4\mathbf{i} - 9\mathbf{j})\ \mathrm{m\,s^{-2}}$
c) $(3\mathbf{i} + 2\mathbf{j})\ \mathrm{m\,s^{-1}},\ (4\mathbf{i} + 9\mathbf{j})\ \mathrm{m\,s^{-2}}$
d) $(2\mathbf{i} - 3\mathbf{j})\ \mathrm{m\,s^{-1}},\ (4\mathbf{i} + 9\mathbf{j})\ \mathrm{m\,s^{-2}}$

37. A particle is projected from a point on a horizontal plane with speed $10\ \mathrm{m\,s^{-1}}$ at an angle 30° to the plane. The horizontal range R covered by the particle is

a) $R = 10\sqrt{3}\ \mathrm{m}$
b) $R = \frac{5}{4}\sqrt{3}\ \mathrm{m}$
c) $R = 5\sqrt{3}\ \mathrm{m}$
d) $R = \frac{5}{2}\sqrt{3}\ \mathrm{m}$

38. When the length of a spring is 60% of its natural length, the thrust in the spring is $10\ \mathrm{N}$, the modulus of elasticity of the spring is

a) $\frac{50}{3}\ \mathrm{N}$
b) $\frac{20}{3}\ \mathrm{N}$
c) $25\ \mathrm{N}$
d) $4\ \mathrm{N}$

39. A centripetal force of $72\ \mathrm{N}$ causes a particle of mass $4\ \mathrm{kg}$ to move in a horizontal circle of radius $2\ \mathrm{m}$. The angular speed of the particle is

a) $9\ \mathrm{rad\,s^{-1}}$
b) $3\ \mathrm{rad\,s^{-1}}$
c) $9\ \mathrm{m\,s^{-1}}$
d) $3\ \mathrm{m\,s^{-1}}$

40. Two particles of masses $2\ \mathrm{kg}$ and $3\ \mathrm{kg}$ are placed at the points $(3, 1)$ and $(8, 6)$ respectively. The position of their center of mass is at the point

a) $(6, 4)$
b) $(5.5, 3.5)$
c) $(5, 3)$
d) $(4, 6)$

41. Fig. 1 shows a uniform sheet of metal in the form of letter L with dimensions in metres.

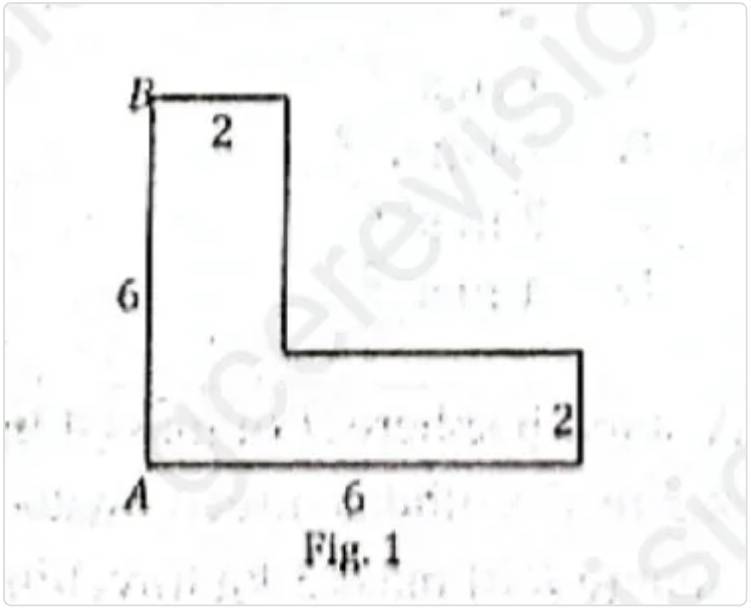


Fig. 1

The distance of its center of mass from edge AB is

a) $2.2\ \mathrm{m}$
b) $1.8\ \mathrm{m}$
c) $3\ \mathrm{m}$
d) $2.5\ \mathrm{m}$

42. A uniform ladder AB, of length $2a$, rests in limiting equilibrium with its top end against a rough vertical wall and its lower end on a rough horizontal floor. The forces acting on the ladder are as shown in fig. 2 below.

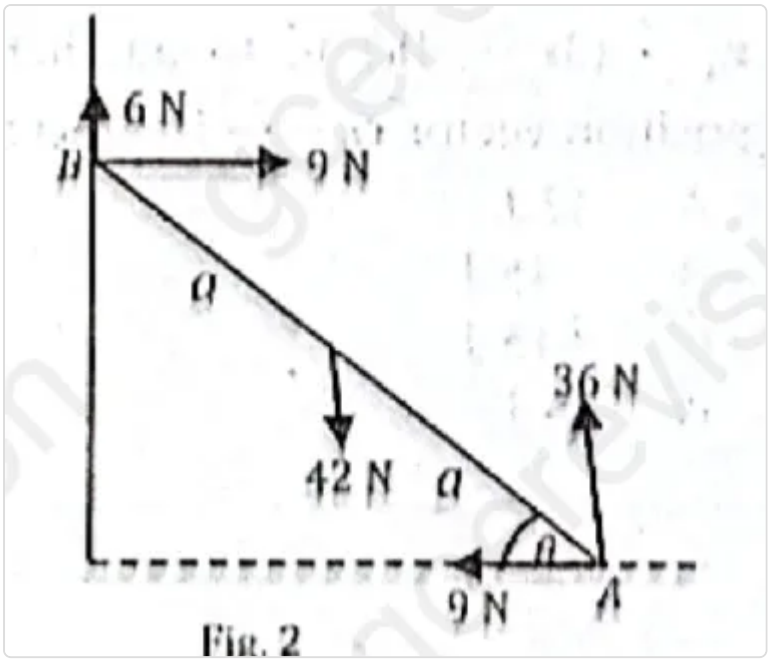


Fig. 2

The angle θ which AB makes with the floor is given by

Options as printed: A) $\tan\theta = \dfrac{3}{5}$ B) $\tan\theta = \dfrac{5}{3}$ C) $\sin\theta = \dfrac{3}{5}$ [as printed] D) $\sin\theta = \dfrac{3}{5}$ [as printed]

43. A girl on a ship, whose velocity is $(4\mathbf{i} - 7\mathbf{j})\ \mathrm{m\,s^{-1}}$ is watching a boat whose velocity is $(-8\mathbf{i} - 2\mathbf{j})\ \mathrm{m\,s^{-1}}$. The speed of the boat as it appears to the girl on the ship is

a) $(-12\mathbf{i} + 5\mathbf{j})\ \mathrm{m\,s^{-1}}$
b) $(12\mathbf{i} - 5\mathbf{j})\ \mathrm{m\,s^{-1}}$
c) $13\ \mathrm{m\,s^{-1}}$
d) $(\sqrt{68} - \sqrt{65})\ \mathrm{m\,s^{-1}}$

44. A particle accelerating from rest has its speed $v\ \mathrm{m\,s^{-1}}$ at time t seconds given by $v = 6t - \dfrac{1}{2}t^2$. The acceleration of the particle when $v = 10$ is

a) $1\ \mathrm{m\,s^{-2}}$
b) $10\ \mathrm{m\,s^{-2}}$
c) $2\ \mathrm{m\,s^{-2}}$
d) $4\ \mathrm{m\,s^{-2}}$

45. A smooth sphere A of mass $4\ \mathrm{kg}$ travelling at $5\ \mathrm{m\,s^{-1}}$ collides directly with a smooth sphere B of mass $3\ \mathrm{kg}$ travelling at $4\ \mathrm{m\,s^{-1}}$ in the opposite direction. Given that the speed of A after impact is $0.5\ \mathrm{m\,s^{-1}}$, the kinetic energy of B after impact is

a) $24\ \mathrm{J}$
b) $3\ \mathrm{J}$

c) $6\ \text{J}$
d) $6.5\ \text{J}$

46. The work done by a force $\mathbf{F} = (\mathbf{i} - 2\mathbf{j} - 4\mathbf{k})\ \text{N}$ which moves its point of application from a point A with position vector $\mathbf{r}_A = (3\mathbf{i} - 4\mathbf{k})\ \text{m}$ to another point B with position vector $\mathbf{r}_B = (-\mathbf{i} + 6\mathbf{k})\ \text{m}$ is

Options as printed: A) $47\ \text{J}$ B) $-35\ \text{J}$ C) $-15\ \text{J}$ D) $35\ \text{J}$

47. The engine of a car of mass $1{,}200\ \text{kg}$ works at a constant rate of $54\ \text{kW}$ up a road inclined at an angle $\sin^{-1}\left(\frac{1}{5}\right)$ to the horizontal. Given that the non-gravitational resistance to the motion of the car is $300\ \text{N}$ and taking g as $10\ \text{m}\,\text{s}^{-2}$, the maximum speed of the car is

a) $20\ \text{m}\,\text{s}^{-1}$
b) $25\ \text{m}\,\text{s}^{-1}$
c) $100\ \text{m}\,\text{s}^{-1}$
d) $900\ \text{m}\,\text{s}^{-1}$

48. A block of mass $4\ \text{kg}$ slides down a rough plane inclined at 30 to the horizontal at constant speed. The block covers a distance $0.8\ \text{m}$. Taking g as $10\ \text{m}\,\text{s}^{-2}$, the work done against friction is

a) $16\ \text{kJ}$
b) $16\ \text{J}$
c) $32\ \text{J}$
d) $32\ \text{kJ}$

49. From a speed of $20\ \text{m}\,\text{s}^{-1}$, a car decelerates uniformly to rest, covering a distance of $80\ \text{m}$. The deceleration of the car is

a) $-\frac{5}{2}\ \text{m}\,\text{s}^{-2}$
b) $\frac{5}{2}\ \text{m}\,\text{s}^{-2}$
c) $8\ \text{m}\,\text{s}^{-2}$
d) $-\frac{2}{5}\ \text{m}\,\text{s}^{-2}$

50. Two dice are thrown together. The probability of having at least a SIX is

a) $\frac{11}{36}$
b) $\frac{1}{18}$
c) $\frac{1}{36}$
d) $\frac{1}{3}$

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